

PHASE-0 STUDENT ENGINEERING PACKET

Behavioral-Intelligence MicroDrone Wildlife Deterrence System Prepared for Rensselaer Polytechnic Institute — September 2026 Prepared by CONTEXTUAL·AI™ SYSTEMS — Jeffrey N. Barat

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SECTION 1 — EXECUTIVE OVERVIEW

Phase-0 Autonomous MicroDrone Wildlife Deterrence System Prepared for Rensselaer Polytechnic Institute — September 2026 Prepared by CONTEXTUAL·AI™ SYSTEMS — Jeffrey N. Barat

The Phase-0 Autonomous MicroDrone Wildlife Deterrence System is a roof-mounted, fully autonomous micro-drone designed to detect approaching birds, classify species, execute species-adaptive deterrence behaviors, and safely return to its dock. The system uses contextual AI, multi-axis motion, dual-sensor detection, and strict safety layers to ensure effective deterrence without risk to humans, animals, or property.

Phase-0 is intentionally lightweight, safe, and student-buildable within one semester. It uses the NVIDIA Jetson Nano as the onboard compute module and incorporates a new Interior Cart Deterrence Mode to prevent habituation when birds enter the cart from the ground.

The Phase-0 system is divided into five engineering modules:

- **Computer Vision Pipeline** — species detection, approach-angle detection, interior-cart detection
- **Motion Engine** — rise, tilt, rotation, hover, interior micro-motion
- **Safety Layer** — human detection, no-launch zones, interior-mode constraints
- **Docking System** — alignment, charging, re-engagement logic
- **Simulation Environment** — Gazebo/Webots for motion-engine testing

RCOS will build and validate the full autonomy system in simulation during the Fall 2026 semester. The physical Phase-0 drone will be built by the MDL team in the following semester using this specification.

SECTION 2 — FULL PHASE-0 SYSTEM SPECIFICATION (Unified & Corrected)

The Phase-0 Autonomous MicroDrone Wildlife Deterrence System is built around a modular architecture that separates perception, intelligence, motion control, safety enforcement, and docking behavior into distinct layers. This ensures clarity for RCOS engineering teams, simplifies debugging, and mirrors real-world robotics design practices.

Below is the complete, corrected Phase-0 specification.

2.1 Core Concept

A roof-mounted autonomous micro-drone that:

- detects approaching wildlife
- classifies species
- selects deterrence patterns
- executes multi-axis motion behaviors
- adapts behavior based on context (exterior vs. interior)
- enforces strict safety rules
- returns to its dock autonomously

Phase-0 is intentionally lightweight, safe, and student-buildable within one semester. After RCOS develops the full autonomy system in simulation, the MDL team will build the physical Phase-0 drone in the following semester using this specification.

2.2 Dual-Sensor Detection Architecture

Exterior Detection

- **Front CSI camera**
 - species classification
 - approach-angle detection
 - multi-bird detection
- **Rear CSI/USB camera**
 - threat direction
 - rotation direction
 - launch priority

Interior Detection

- **Interior bird-entry sensor**
 - detects birds entering the cart from the ground
 - triggers Interior Cart Deterrence Mode

Safety Sensors

- **IMU (I²C)** — tilt, rotation, acceleration
- **IR proximity sensor** — short-range human/collision safety

All sensors feed directly into the Jetson Nano.

2.3 Jetson Nano Compute Layer

The Jetson Nano serves as the onboard AI computer, running:

- Python

- OpenCV
- TensorRT-optimized species classifier
- Motion engine logic
- Safety layer logic
- Docking logic
- GPIO control for motors, sensors, LEDs, and buzzer

The Nano is the central brain of Phase-0.

2.4 Computer Vision Pipeline

The CV pipeline performs:

- species classification
- approach-angle detection
- multi-bird detection
- interior-cart detection
- threat direction analysis
- launch/no-launch decision support

Outputs from CV feed into the Decision Layer.

2.5 Decision Layer

The Decision Layer selects:

- deterrence pattern
- motion parameters
- species-adaptive behavior
- interior vs. exterior mode
- safety overrides
- docking behavior
- battery-aware behavior

This layer is the intelligence core of the system.

2.6 Multi-Axis Motion Engine (Corrected)

Exterior Mode

- rise (1–3 feet)
- tilt (predator posture)
- rotation (full yaw)
- hover
- adaptive pitch/yaw/vertical acceleration

- short-path rotation logic
- **Exterior Close-Buzz Pattern (Cart Unoccupied Only)**
 - short lateral passes near the cart roofline and exterior perimeter
 - limited speed and distance
 - enabled only when the Safety Layer confirms **no humans in or immediately around the cart**
 - automatically disabled if human presence is detected

Interior Mode (No-Rise Rule)

- **no vertical rise**
- micro-yaw oscillation (5–15°)
- micro-tilt pulses (3–8°)
- directed LED/sound cues downward
- strict interior geofence
- randomized patterns
- species-adaptive modulation

Interior mode is intentionally designed with no vertical rise to protect humans and equipment; deterrence inside the cart is achieved through micromotion, light, and sound rather than free flight.

2.7 Species-Adaptive Intelligence Layer

Different species trigger different deterrence patterns:

- **Ravens:** fast, aggressive, deep tilt, rapid escalation
- **Crows:** medium intensity, sweeping pattern
- **Grackles:** jittery, disruptive, fast escalation
- **Gulls:** high hover, circular pattern
- **Ibis/Herons:** gentle, slow, non-aggressive

Interior mode uses scaled-down, randomized versions of these patterns.

2.8 Safety Layer (Corrected)

The Safety Layer overrides all other logic.

Safety Rules

- ignore humans, dogs, golf carts, maintenance vehicles
- no-launch zones
- no-tilt zones
- controlled retreat
- battery-aware behavior

- docking abort logic
- Interior No-Rise Rule
- Human-Presence Rule
- interior motion geofence
- habituation-prevention randomness

Cart-Motion Safety Constraint (New)

If the cart is moving above a small threshold:

- docking is disallowed
- Safety Layer forces Controlled Retreat
- if battery is critical → drone performs emergency landing (no chasing the dock)

This prevents unsafe docking attempts on a moving platform.

2.9 Docking & Re-Engagement Logic (Corrected)

Docking Behavior

- magnetic alignment
- charging contacts
- dock sensor logic
- re-engagement triggers

Docking Motion Constraint (New)

Docking is only attempted when the cart is stationary.

If the cart begins moving during docking:

- **Normal battery:** abort docking → Controlled Retreat
- **Low battery:** abort docking → retreat → wait → reattempt
- **Critical battery:** emergency landing in safest available location (inside cart if possible)

Battery-Aware Docking

- low battery → docking becomes priority
- critical battery → docking abandoned in favor of emergency landing
- normal battery → standard docking sequence

This ensures the drone never “chases” a moving dock and never crashes due to battery exhaustion.

2.10 Architecture Summary Diagram (Text-Based)

Code

[SENSORS]

- Front CSI Camera
- Rear Camera
- Interior Sensor
- IMU
- IR Sensor

↓

[JETSON NANO]

↓

[CV PIPELINE]

↓

[DECISION LAYER]

↓

[MOTION ENGINE] ← Safety Layer (override)

↓

[DOCKING SYSTEM]

2.11 Enhancement Roadmap

- species-adaptive deterrence patterns
- species-adaptive escalation timing
- improved interior-mode modulation
- multi-bird coordination logic

2.12 Market Rationale

Large TAM (total addressable market) in the golf-cart market; ideal for OEM and aftermarket integration.

SECTION 3 — UPDATED BILL OF MATERIALS (WITH ACRONYM EXPLANATIONS)

The Bill of Materials (BOM) lists every component required to build the physical Phase0 MicroDrone Wildlife Deterrence System. All parts are inexpensive, widely available, and chosen specifically so the MDL team can assemble and test the system within one semester.

Note: RCOS will not build hardware during the Fall 2026 semester. RCOS will use these components in simulation only, ensuring that the autonomy software they develop matches the real hardware architecture that MDL will construct in the following semester.

Acronyms are explained so students can understand the hardware without prior robotics experience.

3.1 Core Compute

Jetson Nano 2GB Developer Kit

The Jetson Nano is the onboard AI computer that runs:

- species classification
- motion engine
- safety layer
- docking logic
- GPIO control

Acronym: GPIO — General-Purpose Input/Output (programmable pins used to control motors, sensors, LEDs, and buzzer)

3.2 Sensors

CSI Cameras (Front + Rear)

High-speed cameras used for real-time computer vision. **CSI — Camera Serial Interface**

IMU (Inertial Measurement Unit)

Provides tilt, rotation, and acceleration data. **IC — Inter-Integrated Circuit**

IR Proximity Sensor

Short-range safety sensor used to detect humans or obstacles. **IR — Infrared**

Interior Bird-Entry Sensor

Detects birds entering the cart from the ground and triggers Interior Cart Deterrence Mode.

3.3 Motors & ESCs

Brushless Motors (4× 1103 or 1204)

Provide lift, tilt, and rotation.

ESCs (Electronic Speed Controllers) — 4×

Convert battery power into controlled motor power. **ESC — Electronic Speed Controller** (controls motor speed using PWM signals)

3.4 Power System

LiPo 2S 500–800 mAh Battery

Primary power source for motors and ESCs.

Acronyms:

- **LiPo** — Lithium-Polymer battery
- **2S** — Two LiPo cells in series (~7.4V)
- **mAh** — milliamp-hours (battery capacity)

12V → 5V Buck Converter

Steps down voltage to power the Jetson Nano and dock electronics.

3.5 Deterrence Hardware

LED Strobe

Provides visual deterrence cues (exterior + interior mode).

Piezo Buzzer

Provides audio deterrence cues (exterior + interior mode).

3.6 Docking Hardware

- magnetic alignment rails
- spring-loaded charging contacts
- dock alignment sensor
- weather-resistant enclosure

3.7 Jetson Nano GPIO Connections (Student-Friendly Map)

Sensors

- Front CSI camera → CSI port
- Rear camera → USB or CSI
- IMU → I²C (SDA/SCL)
- IR sensor → GPIO 17
- Interior sensor → GPIO 23

Motors / ESCs

- ESC1 → GPIO 5
- ESC2 → GPIO 6
- ESC3 → GPIO 13
- ESC4 → GPIO 19
- ESC power → LiPo battery

Dock

- dock sensor → GPIO 22
- charging contacts → 5V rail

Deterrence Hardware

- LED → GPIO 24
- buzzer → GPIO 25

Power

- Jetson → 5V buck converter
- ESCs → LiPo battery
- sensors → Jetson 5V rail

SECTION 4 — WIRING & ELECTRONICS GUIDE

This section shows students exactly how to wire the Jetson Nano, sensors, ESCs, motors, deterrence hardware, and the dock. Everything is laid out in simple, readable form so RCOS teams can assemble the Phase-0 system without confusion.

The wiring follows the same GPIO assignments used in Section 3.

4.1 Overview

The wiring system connects:

- Jetson Nano
- cameras
- IMU
- IR sensor
- interior bird-entry sensor
- ESCs + motors

- LED strobe
- piezo buzzer
- dock sensor
- charging contacts
- power system (LiPo + buck converter)

All wiring is designed to be:

- simple
- modular
- easy to debug
- safe for beginners

4.2 Jetson → Sensors

Front CSI Camera

- connect to CSI port
- used for species classification and approach-angle detection

Rear Camera

- connect to USB or CSI
- used for threat direction and rotation logic

IMU (Tilt/Rotation/Acceleration)

- connect via I²C
 - SDA → Jetson SDA
 - SCL → Jetson SCL
- used for motion stability and safety

IR Proximity Sensor

- signal → GPIO 17
- power → Jetson 5V
- ground → Jetson GND
- used for human/collision safety

Interior Bird-Entry Sensor

- signal → GPIO 23
- power → Jetson 5V
- ground → Jetson GND
- triggers Interior Cart Deterrence Mode

4.3 Jetson → ESCs (Motor Control)

Each ESC receives:

- signal from Jetson GPIO
- power directly from LiPo battery
- shared ground with Jetson

ESC Wiring

- ESC1 → GPIO 5
- ESC2 → GPIO 6
- ESC3 → GPIO 13
- ESC4 → GPIO 19

Motor Wiring

Each ESC connects to one brushless motor. Students must ensure:

- correct phase wiring
- secure solder joints
- no exposed metal
- motors spin freely

4.4 Jetson → Docking Hardware

Dock Alignment Sensor

- signal → GPIO 22
- power → Jetson 5V
- ground → Jetson GND

Charging Contacts

- contacts → Jetson 5V rail
- spring-loaded pins ensure reliable charging
- weather-resistant enclosure protects electronics

4.5 Jetson → Deterrence Hardware

LED Strobe

- signal → GPIO 24
- power → Jetson 5V

- ground → Jetson GND

Piezo Buzzer

- signal → GPIO 25
- power → Jetson 5V
- ground → Jetson GND

4.6 Power System Wiring

LiPo Battery (2S, 500–800 mAh)

- powers ESCs + motors
- connects directly to ESC power leads
- must be handled carefully (LiPo safety rules apply)

12V → 5V Buck Converter

- input → LiPo battery
- output → Jetson Nano 5V input
- ensures stable power for compute + sensors

Jetson 5V Rail Powers

- IR sensor
- interior sensor
- dock sensor
- LED strobe
- buzzer

4.7 Wiring Safety Checklist (Student-Friendly)

Students must verify:

- all grounds are shared
- no exposed solder joints
- motors spin freely
- ESCs are correctly mapped to GPIO pins
- sensors receive correct voltage (5V vs. 3.3V)
- buck converter output is stable at 5V
- dock contacts are insulated
- wires are strain-relieved
- no wires interfere with propellers
- LiPo battery is secured and padded

4.8 Quick Wiring Map (Copy-Friendly)

Code

Sensors:

Front CSI → CSI port
Rear Camera → USB/CSI
IMU → I²C (SDA/SCL)
IR Sensor → GPIO 17
Interior Sensor → GPIO 23

Motors / ESCs:

ESC1 → GPIO 5
ESC2 → GPIO 6
ESC3 → GPIO 13
ESC4 → GPIO 19
ESC Power → LiPo

Dock:

Dock Sensor → GPIO 22
Charging Contacts → 5V rail

Deterrence:

LED → GPIO 24
Buzzer → GPIO 25

Power:

Jetson → 5V buck converter
ESCs → LiPo battery
Sensors → Jetson 5V rail

SECTION 5 — SYSTEM ARCHITECTURE DOCUMENT

This section explains how all parts of the Phase-0 MicroDrone Wildlife Deterrence System work together. Students should use this as the “big picture” reference before diving into code or wiring.

The architecture is divided into five major layers:

1. Detection Layer
2. Decision Layer
3. Motion Layer
4. Safety Layer
5. Docking Layer

Each layer has a clear job and communicates with the others through simple, predictable data flows.

5.1 Detection Layer

The Detection Layer gathers all sensor information and sends it to the Jetson Nano.

Inputs

- front CSI camera — species classification, approach-angle detection, multi-bird detection
- rear CSI/USB camera — threat direction, rotation direction, launch priority
- interior bird-entry sensor — detects birds entering the cart
- IMU — tilt, rotation, acceleration
- IR proximity sensor — human/collision safety

Purpose

- provide clean, real-time data to the Decision Layer
- ensure the drone always knows what's happening around it

5.2 Decision Layer

The Decision Layer is the “brain” of the system. It takes sensor data and decides what the drone should do next.

Responsibilities

- select deterrence pattern
- select motion parameters
- choose exterior vs. interior mode
- apply species-adaptive logic
- apply safety overrides
- trigger docking behavior
- trigger battery-aware behavior

Outputs

- commands to the Motion Engine
- safety flags
- docking instructions

5.3 Motion Layer

The Motion Layer physically moves the drone.

Exterior Mode

- rise (1–3 feet)
- tilt (predator posture)
- rotation (full yaw)
- hover
- adaptive pitch/yaw/vertical acceleration
- short-path rotation logic

Interior Mode

- no vertical rise
- micro-yaw oscillation (5–15°)
- micro-tilt pulses (3–8°)
- directed LED/sound cues
- strict interior geofence
- randomized patterns
- species-adaptive modulation

Purpose: execute deterrence behaviors safely and predictably.

5.4 Safety Layer

The Safety Layer overrides everything else. If a safety rule is triggered, it cancels motion commands immediately.

Safety Rules

- ignore humans, dogs, golf carts, maintenance vehicles
- no-launch zones
- no-tilt zones
- controlled retreat
- battery-aware behavior
- docking abort logic
- Interior No-Rise Rule
- Human-Presence Rule
- interior motion geofence
- habituation-prevention randomness

Cart-Motion Safety Constraint (New)

If the cart is moving:

- docking is disallowed
- drone enters Controlled Retreat
- if battery is critical → emergency landing

5.5 Docking Layer

The Docking Layer manages all return-to-dock behavior. It ensures the drone transitions from deterrence or interior mode back to a safe, stable, charging state.

Responsibilities

- detect dock alignment
- manage approach, hover, descent, and contact
- enforce cart-motion constraints
- enforce battery-aware docking rules
- coordinate with the Safety Layer
- maintain stable docked state

Docking Behavior

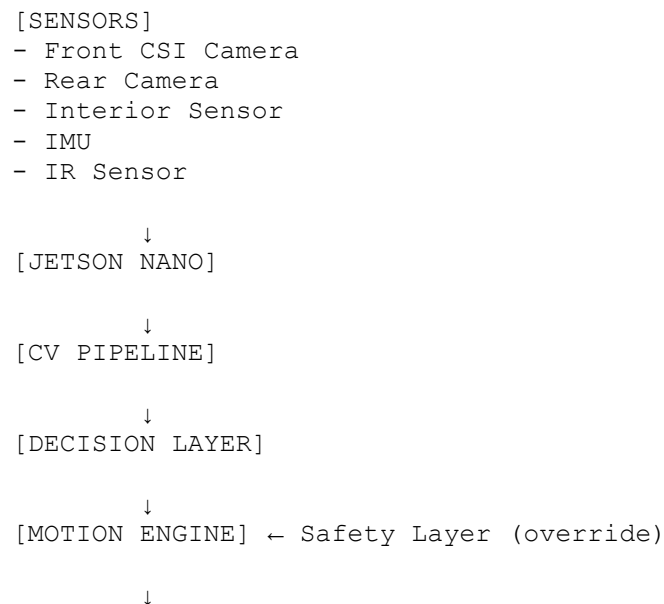
- magnetic alignment
- charging contact engagement
- dock sensor confirmation
- re-engagement triggers when new birds appear

Docking Motion Constraint

Docking is only allowed when the cart is stationary. If the cart moves at any point during docking, the Safety Layer overrides motion and forces a Controlled Retreat.

5.6 Architecture Summary Diagram (Student-Friendly)

Code



This diagram shows the full flow of information and control inside the Phase-0 system.

SECTION 6 — MOTION ENGINE SPECIFICATION

The Motion Engine converts high-level deterrence commands into motor actions using the ESCs. It handles exterior motion, interior micro-motion, stabilization, and safety-aware behavior.

6.1 Motion Engine Overview

The Motion Engine is responsible for:

- vertical lift
- tilt (pitch)
- rotation (yaw)
- hover stability
- interior micro-motion
- species-adaptive modulation
- safety-aware motion limits
- battery-aware motion limits

It receives high-level commands such as:

- “rise 2 feet”
- “tilt 15° forward”
- “rotate left 30°”
- “hover”
- “interior micro-yaw oscillation”

and converts them into ESC signals.

6.2 Exterior Mode Motion Behaviors

Exterior mode is used when the drone is outside the cart and responding to approaching birds.

Rise

- typical rise: 1–3 feet
- smooth acceleration
- IMU-stabilized

Tilt (Predator Posture)

- forward tilt: 10–25°
- stronger tilt for aggressive species

Rotation (Yaw)

- full yaw rotation: 0–360°
- short-path rotation logic chooses fastest turn

Hover

- IMU-corrected
- used before docking and during safety overrides

Exterior Close-Buzz Pattern (Cart Unoccupied Only)

- short lateral passes near the cart roofline and exterior perimeter
- limited speed and distance
- enabled only when the Safety Layer confirms **no humans in or near the cart**
- automatically disabled if human presence is detected
- used to prevent habituation and increase deterrence effectiveness

6.3 Interior Mode Motion Behaviors (No-Rise Rule)

Interior mode is used when a bird enters the cart from the ground.

No Vertical Rise

- drone never lifts off inside the cart
- prevents unsafe propeller motion in confined space

Micro-Yaw Oscillation

- small left-right rotation: 5–15°

Micro-Tilt Pulses

- small forward/backward tilt: 3–8°

Directed LED/Sound Cues

- randomized patterns
- aimed downward

Interior Geofence

- drone remains physically docked
- only micromotion allowed

6.4 Species-Adaptive Motion Patterns

Different birds trigger different motion behaviors.

Ravens

- fast tilt
- rapid oscillation
- strong rotation

Crows

- medium tilt
- sweeping rotation
- moderate oscillation

Grackles

- jittery motion
- rapid micro-tilt pulses
- fast oscillation

Gulls

- higher hover
- circular rotation
- slow tilt

Ibis / Herons

- gentle tilt
- slow oscillation
- non-aggressive patterns

Interior mode uses scaled-down versions of these patterns.

6.5 Safety Overrides (Motion Layer)

The Safety Layer can override the Motion Engine at any time.

Examples

- human detected → cancel motion → hover or stay docked
- cart moving → cancel docking → retreat
- no-launch zone → rise disabled
- no-tilt zone → tilt disabled
- battery-critical → emergency landing
- interior mode → rise disabled, motion limited

Safety always overrides motion commands.

6.6 Battery-Aware Motion Behavior

Normal Battery

- full motion allowed
- exterior mode fully enabled

Low Battery

- reduced motion intensity
- reduced rise height
- reduced tilt angles
- reduced rotation speed
- docking becomes priority

Critical Battery

- deterrence disabled
- docking disabled
- emergency landing
- no rise, no tilt, no rotation
- Safety Layer takes full control

6.7 Motion Engine Control Loop (Student-Friendly)

Code

```
while drone_active:
    read IMU
    read Decision Layer commands
    apply Safety Layer overrides
    compute motor outputs
    send PWM signals to ESCs
    stabilize using IMU feedback
```

PWM — Pulse-Width Modulation (signal used to control motor speed)

6.8 ESC Output Mapping

Each ESC receives a PWM signal:

- ESC1 → front-left motor
- ESC2 → front-right motor
- ESC3 → rear-left motor
- ESC4 → rear-right motor

Motion is created by adjusting the speed of each motor.

6.9 Student Implementation Notes

Students should:

- start with simple rise/tilt/rotate commands
- test IMU stabilization early
- test micro-motion patterns separately
- test safety overrides frequently
- test docking behavior last
- never test interior mode with humans nearby
- secure the LiPo battery before testing
- use simulation before real flight

SECTION 7 — SPECIES-ADAPTIVE INTELLIGENCE SPECIFICATION

The Species-Adaptive Intelligence Layer determines how the drone behaves based on the type of bird approaching or entering the cart. Different species respond to different motion patterns, speeds, and deterrence cues.

7.1 Purpose of Species-Adaptive Intelligence

Different birds require different deterrence strategies:

- **Ravens** — highly intelligent, aggressive → fast, strong motion
- **Crows** — curious, persistent → sweeping patterns
- **Grackles** — jittery → quick micro-motion
- **Gulls** — large, slow → high hover + circular motion
- **Ibis/Herons** — gentle → non-aggressive cues

The Species-Adaptive Layer ensures the drone:

- uses the right pattern for the right species
- escalates only when needed

- stays safe and predictable
- avoids habituation

7.2 Species Classification Pipeline

The Jetson Nano runs a lightweight TensorRT-optimized classifier.

Inputs

- front CSI camera
- rear camera (optional)

Outputs

- species label
- confidence score
- threat direction
- number of birds detected

Student Notes

- no model training required
- pretrained model provided
- students only integrate classifier output

7.3 Exterior Mode Species Behaviors

Ravens

- fast tilt (15–25°)
- rapid oscillation
- strong rotation
- LED + buzzer enabled

Crows

- medium tilt (10–15°)
- sweeping rotation
- moderate oscillation
- LED enabled, buzzer optional

Grackles

- jittery motion

- rapid micro-tilt pulses
- fast oscillation
- LED + buzzer enabled

Gulls

- higher hover (2–3 feet)
- circular rotation
- slow tilt
- LED enabled, buzzer optional

Ibis / Herons

- gentle tilt
- slow oscillation
- LED only

7.4 Interior Mode Species Behaviors (Scaled-Down)

Interior mode uses micro-motion only.

Ravens

- micro-yaw: 10–15°
- micro-tilt: 5–8°
- LED + buzzer randomized

Crows

- micro-yaw: 8–12°
- LED randomized
- buzzer optional

Grackles

- quick micro-tilt pulses
- fast micro-yaw oscillation
- LED + buzzer randomized

Gulls

- slow micro-yaw: 5–10°
- LED only

Ibis / Herons

- very gentle micro-motion
- LED only

7.5 Escalation Logic

Escalation Triggers

- bird continues approaching
- bird ignores first deterrence pattern
- multiple birds detected
- high confidence species match
- threat direction confirmed

Escalation Steps

1. mild motion + LED
2. stronger motion + LED
3. full motion + LED + buzzer
4. safety override if bird gets too close

Interior Mode Escalation

- micro-yaw
- micro-tilt
- LED + buzzer randomized
- safety override if humans detected

7.6 Randomization (Anti-Habituation)

Birds can learn patterns. To prevent this, the drone randomizes:

- timing
- tilt angles
- yaw angles
- LED patterns
- buzzer patterns
- micro-motion sequences

Example

Code

```
yaw_angle = random(5°, 15°)
tilt_angle = random(3°, 8°)
pattern_delay = random(0.2s, 0.6s)
```

7.7 Species-Adaptive Decision Flow

Code

```
if species == "raven":
    use raven pattern
elif species == "crow":
    use crow pattern
elif species == "grackle":
    use grackle pattern
elif species == "gull":
    use gull pattern
elif species in ["ibis", "heron"]:
    use gentle pattern
else:
    use default pattern
```

Interior mode uses the same logic but with micro-motion only.

7.8 Student Implementation Notes

Students should:

- start with simple species mapping
- test each species pattern separately
- test interior mode patterns last
- verify randomization
- verify escalation
- verify safety overrides
- never test buzzer indoors
- always test with propellers removed first

SECTION 8 — SAFETY LAYER SPECIFICATION

The Safety Layer is the most important part of the Phase-0 system. It overrides motion, species logic, deterrence, and docking to ensure the drone never behaves in a way that could harm people, property, or itself.

8.1 Purpose of the Safety Layer

The Safety Layer ensures:

- the drone never launches at the wrong time
- the drone never tilts or rotates near humans
- the drone never rises inside the cart

- the drone never docks while the cart is moving
- the drone never crashes due to low battery
- the drone always retreats safely when needed

It is designed to be simple, predictable, and easy for students to test.

8.2 Safety Inputs

The Safety Layer monitors:

- IMU
- IR proximity sensor
- interior bird-entry sensor
- dock alignment sensor
- battery level
- cart-motion state
- Motion Engine commands
- Species-Adaptive commands
- docking commands

If any input indicates danger, the Safety Layer overrides the system immediately.

8.3 Safety Rules (Student-Friendly)

Rule 1 — Human Presence Rule If a human is detected nearby:

- cancel all motion
- cancel tilt
- cancel rotation
- cancel deterrence
- drone stays docked or hovers safely
- buzzer disabled indoors

Rule 2 — No-Launch Zones If the drone is inside the cart:

- rise disabled
- launch disabled
- only micromotion permitted

Rule 3 — No-Tilt Zones If humans or objects are close:

- tilt disabled
- rotation limited

Rule 4 — Interior No-Rise Rule Inside the cart:

- rise always disabled
- drone stays docked

Rule 5 — Controlled Retreat If anything unsafe happens:

- cancel deterrence
- hover briefly
- retreat to safe position

Rule 6 — Battery-Aware Safety Low battery → reduced motion Critical battery → emergency landing

Rule 7 — Docking Safety Docking allowed only when:

- cart stationary
- battery not critical
- no humans nearby

Rule 8 — Cart-Motion Safety Constraint If the cart is moving:

- docking disallowed
- drone enters Controlled Retreat

Rule 9 — Habituation Prevention Randomize LED, buzzer, micromotion

★ **Rule 10 — Exterior Close-Buzz Rule (New)**

Exterior close-buzz behavior is allowed only when the cart is unoccupied.

If no humans are detected in or immediately around the cart:

- drone may perform short, controlled passes near the cart roofline and exterior perimeter
- motion intensity remains limited and IMU-stabilized
- used to prevent habituation and increase deterrence effectiveness

If any human is detected:

- close-buzz behavior is immediately disabled
- drone reverts to standard exterior deterrence (rise/tilt/rotation)
- interior mode remains micromotion only

SECTION 9 — DOCKING SYSTEM SPECIFICATION

The Docking System ensures the drone can safely return to its charging station after deterrence, interior mode activity, or safety overrides.

9.1 Purpose of the Docking System

The docking system allows the drone to:

- return safely after deterrence
- recharge its battery
- re-engage when new birds approach
- stay stable during interior mode
- avoid unsafe docking attempts
- handle cart-motion correctly
- perform emergency landing if battery-critical

Docking is one of the most important safety-critical behaviors in Phase-0.

9.2 Docking Hardware Components

- **Magnetic alignment rails** — guide the drone into position
- **Spring-loaded charging contacts** — ensure reliable electrical contact
- **Dock alignment sensor** — confirms correct positioning
- **Weather-resistant enclosure** — protects electronics
- **Jetson GPIO connections:**
 - dock sensor → GPIO 22
 - charging contacts → 5V rail

9.3 Docking Detection Logic

The drone uses:

- rear camera (optional)
- IMU
- dock alignment sensor
- battery level
- cart-motion state

to determine when docking is safe.

Docking Allowed When

- cart stationary
- battery not critical
- dock sensor stable
- no humans nearby

- drone in exterior mode
- Safety Layer approves docking

Docking Disallowed When

- cart moving
- battery critical
- humans nearby
- interior mode active
- Safety Layer blocks docking

9.4 Docking Motion Sequence (Student-Friendly)

1. **Approach Dock** — drone moves toward dock slowly
2. **Hover Over Dock** — IMU stabilizes position
3. **Alignment Check** — dock sensor confirms alignment
4. **Slow Descent** — drone lowers gently
5. **Contact Detection** — charging contacts engage
6. **Power Transfer** — Jetson switches to dock power
7. **Docked State** — drone enters low-power mode

9.5 Docking Motion Constraint (New)

Docking is only attempted when the cart is stationary.

This prevents:

- misalignment
- crashes
- battery-critical failures
- unsafe landings
- chasing a moving dock

9.6 Cart-Motion Docking Behavior (New)

If the cart begins moving before docking

- drone waits
- drone hovers safely
- drone enters Controlled Retreat
- docking postponed until cart stops

If the cart begins moving during docking

Normal Battery

- abort docking
- hover briefly
- Controlled Retreat
- reattempt when cart stops

Low Battery

- abort docking
- retreat
- wait for cart to stop
- reattempt immediately

Critical Battery

- docking disabled
- emergency landing
- no chasing the dock
- land in safest available location

9.7 Battery-Aware Docking Logic

Normal Battery

- docking allowed
- full motion available
- standard sequence

Low Battery

- docking becomes priority
- reduced motion intensity
- reduced tilt/rotation
- dock as soon as cart stops

Critical Battery

- docking disabled
- deterrence disabled
- emergency landing
- Safety Layer takes full control

9.8 Docked State Behavior

When docked:

- drone enters low-power mode
- LED and buzzer disabled
- motors off
- sensors remain active
- interior micro-motion allowed
- drone waits for new detections
- drone re-engages automatically

9.9 Student Implementation Notes

Students should:

- test docking with propellers removed
- test alignment sensor first
- test slow descent carefully
- test cart-motion abort logic by gently pushing the cart
- test battery-critical behavior in simulation
- verify emergency landing logic
- verify charging contacts
- secure wires inside dock enclosure
- never test docking indoors with people nearby

SECTION 10 — INTERIOR CART DETERRENCE MODE SPECIFICATION

Interior Cart Deterrence Mode activates when a bird enters the golf cart from the ground. This mode is completely different from exterior deterrence because the drone must remain physically docked, avoid humans, and use only micro-motion and light/sound cues.

This section explains how interior mode works, how it stays safe, how it adapts to species, and how it prevents habituation.

10.1 Purpose of Interior Mode

Interior mode exists to:

- deter birds that enter the cart from the ground
- keep the drone safely docked
- avoid collisions with humans or objects
- use gentle, controlled cues
- prevent habituation through randomness

- maintain safety at all times

Interior mode is never aggressive and never uses full motion.

10.2 Interior Mode Activation

Interior mode activates when:

- the interior bird-entry sensor detects movement inside the cart
- the drone is already docked
- no exterior deterrence is running
- the Safety Layer approves interior mode

Interior mode does **not** activate if:

- humans are inside the cart
- battery is critical
- cart is moving
- drone is in the middle of docking
- Safety Layer blocks motion

10.3 Interior Mode Safety Rules

Interior mode follows strict safety rules:

Rule 1 — No-Rise Rule

The drone never lifts off inside the cart.

Rule 2 — Docked-Only Motion

All motion occurs while the drone remains physically docked.

Rule 3 — Human-Presence Rule

If humans are detected:

- motion disabled
- buzzer disabled
- LED randomized only
- drone stays docked

Rule 4 — Micro-Motion Only

Motion is limited to:

- micro-yaw oscillation
- micro-tilt pulses
- small LED/buzzer cues

Rule 5 — Interior Geofence

Motion angles strictly limited:

- yaw: 5–15°
- tilt: 3–8°

Rule 6 — Habituation Prevention

Patterns are randomized so birds cannot learn them.

10.4 Interior Mode Motion Behaviors

Interior mode uses micro-motion to create deterrence without leaving the dock.

Micro-Yaw Oscillation

- small left-right rotation
- range: 5–15°
- slow, smooth movement
- creates visual motion cues

Micro-Tilt Pulses

- small forward/backward tilt
- range: 3–8°
- short pulses
- adds unpredictability

LED Cues

- directed downward
- randomized brightness
- randomized timing
- used for all species

Buzzer Cues

- used only when no humans are present
- randomized chirps
- never loud

- never continuous

10.5 Species-Adaptive Interior Patterns

Interior mode uses scaled-down versions of exterior patterns.

Ravens

- micro-yaw: 10–15°
- micro-tilt: 5–8°
- LED + buzzer randomized

Crows

- micro-yaw: 8–12°
- LED randomized
- buzzer optional

Grackles

- quick micro-tilt pulses
- fast micro-yaw oscillation
- LED + buzzer randomized

Gulls

- slow micro-yaw: 5–10°
- LED only
- no buzzer

Ibis / Herons

- very gentle micro-motion
- LED only
- no buzzer

10.6 Interior Mode Escalation

Interior escalation is gentle and controlled.

Escalation Steps

1. micro-yaw
2. micro-tilt pulses

3. LED randomized
4. LED + buzzer randomized (only if no humans present)

Safety Override

If humans appear:

- motion stops
- buzzer stops
- LED only
- drone stays docked

10.7 Interior Mode Randomization (Anti-Habituation)

Birds can learn patterns, so interior mode uses small random variations:

Code

```
yaw_angle      = random(5°, 15°)
tilt_angle     = random(3°, 8°)
pattern_delay  = random(0.2s, 0.6s)
led_brightness = random(30%, 70%)
buzzer_chirp   = random(50ms, 150ms)
```

Randomness is controlled and safe.

10.8 Interior Mode Deactivation

Interior mode ends when:

- bird leaves the cart
- interior sensor clears
- humans enter the cart
- battery becomes low or critical
- cart begins moving
- docking sequence begins
- Safety Layer overrides motion

When interior mode ends:

- drone returns to docked idle state
- LED and buzzer turn off
- sensors remain active
- drone waits for new detections

10.9 Student Implementation Notes

Students should:

- test interior mode with propellers removed
- verify micro-motion angles carefully
- test LED patterns first
- test buzzer only outdoors
- test human-presence override frequently
- test randomization logic
- test interior sensor reliability
- never test interior mode with people sitting in the cart
- ensure drone remains fully docked during all tests

SECTION 11 — BUILD PLAN

SECTION 11 — BUILD PLAN

This Build Plan describes the physical construction of the Phase-0 MicroDrone Wildlife Deterrence System. These steps outline how the MDL team will assemble the drone, integrate the Jetson Nano, connect sensors, mount the docking hardware, and perform initial physical validation.

Note: Sections 11.1–11.9 describe the physical construction of the Phase-0 drone, which will be performed by the MDL team in the semester following RCOS’s autonomy development. RCOS will complete software, simulation, and validation only.

11.1 Build Sequence Overview

1. Frame assembly
2. Motor + ESC installation
3. Jetson Nano mounting
4. Sensor installation (CSI cameras, IMU, IR sensor, interior sensor)
5. Docking hardware installation
6. Wiring and power system integration
7. Initial bench testing
8. Software load (RCOS autonomy stack)
9. Physical validation and tuning

11.2 MDL Responsibilities

The MDL team is responsible for all physical construction and integration, including:

- assembling the drone frame
- installing motors and ESCs
- mounting the Jetson Nano

- wiring sensors and deterrence hardware
- installing and weatherproofing the dock
- integrating the power system
- performing physical testing and tuning
- loading and validating the RCOS autonomy software on real hardware

11.3 RCOS Deliverables Required Before MDL Build

Before MDL begins physical construction, RCOS will deliver:

- Computer Vision pipeline
- Motion Engine logic
- Safety Layer logic
- Docking logic
- Simulation validation results
- full codebase and documentation
- species-adaptive behavior modules
- interior-mode logic
- battery-aware behavior logic

These deliverables ensure the MDL team can load the completed autonomy stack onto the Jetson Nano and begin physical integration.

11.4 Phase 4 — Dock Construction

Note: Dock construction is performed by the MDL team. RCOS does not build physical hardware.

Step 1 — Build Dock Base

- use 3D-printed or laser-cut parts
- ensure stable mounting on cart roof

Step 2 — Install Magnetic Rails

- align rails so drone centers automatically

Step 3 — Install Charging Contacts

- spring-loaded pins
- connect to Jetson 5V rail
- ensure good contact pressure

Step 4 — Install Dock Sensor

- connect to GPIO 22
- test alignment detection

Step 5 — Weatherproofing

- seal enclosure
- protect electronics

11.5 Phase 5 — Motor & ESC Installation

Note: Sections 11.5–11.9 describe MDL-only physical build steps.

Step 1 — Mount Motors

- secure 1103/1204 brushless motors
- ensure free rotation and no wire interference

Step 2 — Install ESCs

- mount ESCs near motors
- ensure airflow and heat dissipation

Step 3 — Connect ESC Power Leads

- connect directly to LiPo battery
- verify polarity and insulation

Step 4 — Connect ESC Signal Wires

- ESC1 → GPIO 5
- ESC2 → GPIO 6
- ESC3 → GPIO 13
- ESC4 → GPIO 19

Step 5 — Bench Test Motor Spin

- test each motor individually
- verify correct direction and mapping

11.6 Phase 6 — Sensor Installation

Step 1 — Install Front CSI Camera

- mount securely
- connect to CSI port

Step 2 — Install Rear Camera

- mount facing backward
- connect to USB or CSI

Step 3 — Install IMU

- mount flat and vibration-isolated
- connect via I²C (SDA/SCL)

Step 4 — Install IR Proximity Sensor

- mount facing outward
- connect to GPIO 17

Step 5 — Install Interior Bird-Entry Sensor

- mount inside cart roofline
- connect to GPIO 23

11.7 Phase 7 — Wiring & Power Integration

Step 1 — Install Buck Converter

- input → LiPo battery
- output → Jetson 5V rail

Step 2 — Connect Jetson Power

- verify stable 5V output
- check polarity

Step 3 — Connect Sensors to Jetson 5V Rail

- IR sensor
- interior sensor
- dock sensor
- LED strobe
- buzzer

Step 4 — Strain-Relief All Wires

- zip-tie bundles
- avoid propeller interference

11.8 Phase 8 — Deterrence Hardware Installation

Step 1 — Install LED Strobe

- mount facing downward
- connect to GPIO 24

Step 2 — Install Piezo Buzzer

- mount inside enclosure
- connect to GPIO 25

Step 3 — Test LED/Buzzer Output

- verify correct GPIO mapping
- verify intensity and timing

11.9 Phase 9 — Physical Validation & Tuning

Step 1 — Load RCOS Autonomy Stack

- install software on Jetson Nano
- verify boot and runtime stability

Step 2 — Test Sensor Inputs

- verify camera frames
- verify IMU readings
- verify IR and interior sensor triggers
- verify dock sensor alignment

Step 3 — Test Motion Engine

- test rise, tilt, rotation
- test interior micromotion
- test exterior close-buzz (cart unoccupied only)

Step 4 — Test Safety Layer

- human-presence overrides
- no-launch zones
- no-tilt zones
- battery-aware behavior
- cart-motion docking constraints

Step 5 — Test Docking Behavior

- approach
- hover
- alignment
- descent
- charging contact engagement

Step 6 — Final Tuning

- adjust motion parameters
- adjust deterrence patterns
- adjust sensor thresholds

SECTION 12 — TESTING & VALIDATION PLAN

SECTION 12 — TESTING & VALIDATION PLAN

This Testing & Validation Plan describes how the Phase-0 system will be verified for correctness, safety, and performance.

RCOS will perform all testing in simulation during the Fall 2026 semester. The detailed physical testing steps below (12.1–12.5) will be performed by the MDL team once the Phase-0 drone is built in the following semester. This ensures the autonomy software developed by RCOS is fully validated in simulation before MDL performs real-world testing on the physical drone.

12.1 Phase 1 — Bench Testing (Propellers Off)

Bench testing verifies that all electronics, sensors, and safety systems function correctly before any motorized motion is attempted.

Test 1 — Power System

- verify buck converter outputs stable 5V
- verify Jetson boots correctly
- verify ESCs receive power
- verify sensors power on

Test 2 — Camera Check

- confirm front CSI camera feed
- confirm rear camera feed
- verify orientation
- test low-light performance

Test 3 — IMU Check

- verify tilt readings
- verify rotation readings
- verify acceleration readings
- confirm smooth response

Test 4 — IR Sensor Check

- move hand in front of sensor

- confirm detection
- confirm safety override

Test 5 — Interior Sensor Check

- simulate bird entry
- confirm interior mode activation
- confirm micro-motion commands

Test 6 — Dock Sensor Check

- place drone in dock
- confirm alignment detection
- confirm docked state logic

Test 7 — LED + Buzzer Check

- test LED brightness
- test LED patterns
- test buzzer chirps (outdoors only)

12.2 Phase 2 — Motion Testing (Propellers Off)

Motion testing verifies that the Motion Engine responds correctly to commands while the drone remains safely grounded.

Test 1 — Rise Command

- motors spin up
- drone stays grounded
- IMU stabilizes

Test 2 — Tilt Command

- motors adjust speed
- IMU shows correct tilt

Test 3 — Rotation Command

- motors adjust yaw
- IMU shows correct rotation

Test 4 — Hover Logic

- motors stabilize

- IMU shows stable readings

Test 5 — Safety Overrides

- IR sensor → motion cancels
- interior sensor → rise disabled
- cart-motion simulation → docking aborts
- battery-critical simulation → emergency landing logic

Test 6 — Interior Mode Micro-Motion

- micro-yaw oscillation
- micro-tilt pulses
- LED patterns
- buzzer patterns (outdoors only)

12.3 Phase 3 — Exterior Flight Testing (Propellers On)

Exterior flight testing verifies real-world motion behavior, stabilization, and safety overrides.

Safety Requirements

- outdoor area
- no people nearby
- battery fully charged
- drone tether available
- eye protection
- LiPo safety bag

Test 1 — Controlled Lift

- drone rises 1–2 feet
- IMU stabilizes
- drone hovers smoothly

Test 2 — Tilt + Rotation

- test predator tilt
- test yaw rotation
- test short-path rotation logic

Test 3 — Species-Adaptive Patterns

Simulate species using software triggers:

- raven
- crow
- grackle
- gull
- ibis/heron

Test 4 — Safety Overrides

- human-presence override
- no-tilt override
- no-launch override
- battery-critical override
- cart-motion override

Test 5 — Controlled Retreat

- trigger retreat
- drone backs away safely
- drone stabilizes and waits

12.4 Phase 4 — Docking + Interior Mode Testing (Propellers On)

Docking and interior-mode testing verify safe transitions between deterrence, docking, and interior micromotion.

Test 1 — Dock Approach

- drone approaches slowly
- hover stabilizes above dock
- alignment sensor confirms position

Test 2 — Dock Descent

- drone lowers gently
- charging contacts engage
- Jetson switches to dock power

Test 3 — Cart-Motion Abort

- push cart slightly
- drone aborts docking
- drone retreats safely
- drone reattempts when cart stops

Test 4 — Low-Battery Docking

- simulate low battery
- drone prioritizes docking
- drone docks as soon as cart stops

Test 5 — Critical-Battery Emergency Landing

- simulate critical battery
- drone cancels docking
- drone performs emergency landing
- drone does not chase moving dock

Test 6 — Interior Mode (Real Conditions)

- simulate bird entering cart
- drone stays docked
- micro-motion activates
- LED cues activate
- buzzer cues activate (if no humans present)
- safety overrides verified

12.5 Student Testing Notes

Students should:

- always test with propellers removed first
- never test buzzer indoors
- never test rise inside the cart
- always secure LiPo battery
- test safety overrides frequently
- test interior mode last
- test docking only outdoors
- test emergency landing in a safe area
- document all results
- report unexpected behavior immediately

SECTION 13 — GLOSSARY OF TERMS & ACRONYMS

This glossary explains all acronyms, technical terms, and robotics vocabulary used throughout the Phase-0 MicroDrone Wildlife Deterrence System packet.

A

AI — Artificial Intelligence Software that makes decisions or classifications based on data.

Alignment Sensor Detects when the drone is correctly positioned in the dock.

B

Battery-Critical Battery level so low that the drone must perform an emergency landing.

Battery-Low Battery level where the drone must reduce motion and prioritize docking.

Buck Converter Device that reduces voltage (e.g., 12V $\rightarrow$ 5V).

Buzzer (Piezo Buzzer) Small speaker used for deterrence chirps.

C

Cart-Motion Constraint Safety rule preventing docking while the golf cart is moving.

CSI — Camera Serial Interface High-speed camera connection used by the Jetson Nano.

Controlled Retreat Safe backward movement used when docking or deterrence must be aborted.

CV = Computer Vision.

It refers to the entire perception pipeline running on the Jetson Nano:

- species detection
- approach-angle detection
- interior-cart detection
- threat direction
- multi-bird detection
- launch/no-launch decision support

It is one of the five Phase-0 engineering modules:

1. **Computer Vision (CV)**
2. Motion Engine
3. Safety Layer
4. Docking System
5. Simulation Environment

D

Deterrence Pattern Motion, light, or sound behavior used to scare away birds.

Dock Charging station mounted on the golf cart roof.

Docked State Drone is physically seated in the dock and running on dock power.

E

ESC — Electronic Speed Controller Controls motor speed using PWM signals.

Emergency Landing Forced landing triggered by critical battery or safety override.

Exterior Mode Drone behavior when operating outside the cart.

F

Front CSI Camera Primary camera used for species classification and threat detection.

G

Geofence (Interior Geofence) Strict limit on motion angles inside the cart to prevent collisions.

GPIO — General-Purpose Input/Output Programmable pins on the Jetson Nano used to control sensors and motors.

H

Habituation When birds learn and ignore repeated patterns. Randomization prevents this.

Hover Drone maintains stable position using IMU feedback.

I

I²C — Inter-Integrated Circuit Two-wire communication bus used for sensors like the IMU.

IMU — Inertial Measurement Unit Measures tilt, rotation, and acceleration.

Interior Mode Drone behavior when a bird enters the cart from the ground.

IR — Infrared Light outside the visible spectrum used for proximity detection.

L

LED Strobe Light used for deterrence cues.

LiPo — Lithium-Polymer Battery Lightweight, high-power battery used in drones.

M

Micro-Motion Small, safe movements used in interior mode (micro-yaw, micro-tilt).

Motion Engine Software that converts commands into motor actions.

N

No-Rise Rule Interior safety rule preventing the drone from lifting off inside the cart.

P

PWM — Pulse-Width Modulation Signal used to control motor speed.

Propellers (Props) Rotating blades that generate lift.

R

Randomization Small variations in timing, angles, and cues to prevent habituation.

Rear Camera Camera used for threat direction and rotation logic.

S

Safety Layer Software that overrides all other behavior to prevent unsafe actions.

Species-Adaptive Intelligence Logic that adjusts deterrence patterns based on bird species.

T

TensorRT NVIDIA library used to accelerate AI models on the Jetson Nano.

Tilt Forward/backward angle of the drone.

Y

Yaw Left-right rotation of the drone.